

# Rakuten Viki: Accelerating innovation in entertainment with Google Cloud

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## ABOUT RAKUTEN VIKI

Rakuten Viki moved its managed service environment on Google Cloud to Kubernetes Engine, which frees up time for the company to focus on delivering innovation.



## About Rakuten Viki

Rakuten Viki is a premier destination for Asian entertainment, with a global audience that comes for TV shows, movies, and Viki Originals, translated into more than 200 languages by a community of avid fans. Reaching more than 40 million fans in more than 190 countries, the company also develops and invests in Viki Originals, such as the award-winning series, Dramaworld.

**Google  
Cloud  
results**

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**Reduced  
managed**

Industries: Media & Entertainment

- Supports Viki's rapid user growth over the last year by supporting a 2x increase in traffic without needing additional provisioning
- Increases the number of concurrent users on Soompi website from 8,000 to 30,000 users
- Achieves sub-second page interaction time and 99.9% uptime

## operational development overhead

Location: Singapore

(<https://cloud.google.com/container-registry/>),

Google Cloud (<https://cloud.google.com/>),

Kubernetes Engine

App Engine (<https://cloud.google.com/appengine/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Build (<https://cloud.google.com/build/>)

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Container Registry

(<https://cloud.google.com/container-registry/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

More than 40 million fans visit Rakuten Viki

(<https://www.viki.com/?locale=en>) every month to

watch their favorite Asian TV shows, movies, and

original content. The company relies heavily on its people and technology to deliver the best possible user experience across different devices and platforms.

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—Alex Chan, Senior Vice President of Product and Engineering, Rakuten Viki

Previously, Rakuten Viki ran its applications and backend services such as its content management system and subscription management on a “bare metal” environment that comprised actual hardware. But the company was experiencing exceptional growth and needed to find a technology platform to support it. Servers were over provisioned to handle traffic spikes, and the engineers were bogged down by infrastructure management. In 2018, Rakuten Viki decided to move its technology stack to the cloud. The company considered various cloud providers but eventually chose Google Cloud Platform (<https://cloud.google.com/>) for its network performance and scaling capabilities.

Rakuten Viki moved 30 microservices that make up its backend API and data-driven business logic to a fully managed environment on Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>) (GKE) for optimal efficiency. The company has experienced exponential growth in recent years. Its subscriber base grew more than 10 times over between 2015 and 2019.

“Google Cloud offers platform as a service that enables us to free up engineering resources from managing infrastructure so we can focus on scaling the business to serve rapidly growing user demand,” says Alex Chan, Senior Vice President of Product and Engineering at Rakuten Viki. “Using autoscaling in GKE, we can scale clusters up and down based on user demand with zero intervention. Instead of overprovisioning, we pay for what we use.”

By running workloads on Google Cloud, Rakuten Viki reduces the cost of provisioning hundreds of servers and the time it takes for the engineers to manage the servers across multiple regions. The company now streams billions of minutes per month on Google Cloud with minimal downtime. On top of that, the Kubernetes clusters are fully managed by Google Site Reliability Engineers to ensure they are available and up to date.

## **Improving workflows with Google Cloud**

A lean engineering team supports infrastructure and development at Rakuten Viki. The engineers are responsible for improving user experience as well as building and deploying new features as quickly as possible. Before moving to Google Cloud, Rakuten Viki depended on a handful of engineers who had specific server knowledge to resolve network issues.

Alex adds, “We need to break down siloed IT functions by getting more people up to speed on the infrastructure without building specialized skills. On Google Cloud, our developers are empowered to change codes and configuration settings with the click of a button to get their work done.”

For example, if the company needs to scale up to cope with more people coming on the system or to stress test a new application, authorized users can modify settings by selecting the Advanced Edit button. Engineers no longer have to put their work on hold while they wait for an administrator to perform infrastructure maintenance.

“Our engineers used to spend a lot of time and effort on server management,” says Omkiran Sharma, Director of Engineering at Rakuten Viki. “Cloud SQL (<https://cloud.google.com/sql/>) on Google Cloud automates all our backups, replication, patches, and updates so our engineering team can focus on delivering innovation.”

## **Building a robust entertainment news website with App Engine**

“We decided to use Google Kubernetes Engine to deploy microservices in order to minimize service disruptions with the help of autoscaling. Ever since we moved to Google Cloud, we can handle bursty events very easily.”

*—Omkiran Sharma,  
Director of Engineering,  
Rakuten Viki*

As a digital content provider, Rakuten Viki requires high network performance to deliver 50,000 hours of entertainment content to millions of users across the world. Traffic to its Soompi website, the world's largest English online media platform for Korean entertainment news, lulls and peaks throughout the day as fans log on from different countries and time zones. Poor website performance not only affects content viewing but also results in user complaints.

"We decided to use Google Kubernetes Engine to deploy microservices in order to minimize service disruptions with the help of autoscaling," Omkiran shares. "Ever since we moved to Google Cloud, we can handle bursty events very easily."

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*—Alex Chan, Senior Vice President of Product and Engineering, Rakuten Viki*

Before the migration, the engineering team ran a stress test to see how the new Soompi site would perform when hit with traffic spikes. Using a load-testing tool, Rakuten Viki simulated 30,000 users visiting the website in a short time. Soompi was able to successfully handle traffic four times more than its normal peak and loaded within 200 milliseconds.

"At Rakuten Viki, our business comprises multiple properties, applications, and backend services. We need to optimize for agility and consider the best technology to implement for each use case," says



Alex. “For example, if we’re trying out something new, it’s simpler to put it on [App Engine](https://cloud.google.com/appengine/) (https://cloud.google.com/appengine/). If it scales up or requires dependencies, we put it on GKE.”

## **Guarding security with Cloud Identity-Aware Proxy**

Rakuten Viki is keenly aware of the growing security threats targeting online businesses and takes proactive measures to protect its IT infrastructure and customer data. Google Cloud helps Rakuten Viki to secure against data leaks by encrypting customer content at rest and in transit.

“We include security by default when building our applications and use many of the robust security and identity features offered by Google Cloud,” Omkiran comments. “Since we’re [Google Workspace](https://workspace.google.com/) (https://workspace.google.com/) users, we configure Security Assertion Markup Language (SAML) to authenticate employees and use [Cloud Identity-Aware Proxy](https://cloud.google.com/iap/) (https://cloud.google.com/iap/) (Cloud IAP) to maintain access control.”

Cloud Identity-Aware Proxy verifies user identity and the context of the request to determine if a user should be granted access to applications and VMs deployed on Google Cloud.

## **Looking toward future growth with Google Play and Android**

Moving forward, Rakuten Viki is growing its business and global footprint through [Google Play](#).

(<https://play.google.com/>) and Android. Partnering with Google over the past few years, the company refined its “Instant App” experience, where users can enjoy the app without fully downloading it, and launched many new features. Now Rakuten Viki is seeing encouraging growth across Asia, Europe, Latin America, and North America. The team is more motivated than ever to deliver happiness to its users and bring the world closer together through engaging content from Asia.

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see how  
Google Cloud  
can help your

business.

